

Grammar or Feeling? Factorized Non-Manual Modeling for Emotion-Aware Sign Language Translation on Commodity Hardware

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Abstract—[TODO: 180–220 words. Follow the 5-sentence recipe in PAPER_TEMPLATE.md: (1) gap: systems model hands, discard the grammatically-obligatory and affect-bearing face; (2) insight: grammatical and affective non-manual function can be explicitly factorized, and manual prosody belongs to affect; (3) method: two-branch encoder with gradient reversal, orthogonality and MI minimization, feeding emotion-conditioned text generation and an expressive avatar; (4) result: XX.X wF1 on EmoSign under leave-one-signer-out CV vs 20.76 for GPT-4o from video; (5) efficiency: end-to-end in XX.X MB VRAM at XX.X ms p95 on an RTX 3050, versus the 80 GB A100 used for the baseline evaluations.]

Index Terms—sign language translation, non-manual markers, emotion recognition, affective computing, disentangled representations, expressive avatars, efficient inference

I. INTRODUCTION

[TODO: P1 – the stakes. Sign language is a full language; affect-free translation strips the person from the message. Cite documented consequences as motivation, not as a deployment pitch.]

[TODO: P2 – the technical gap. Non-manual signals are grammatically obligatory (brow raise marks yes/no questions, brow furrow marks wh-questions, head-shake marks negation, mouth morphemes modify predicates) [1], [2] and simultaneously affective [3]. Affective prosody in ASL is carried partly by the hands [4], [5].]

[TODO: P3 – the hook. Hearing non-signers systematically misread linguistic markers as negative emotion, and so do frontier models: on EmoSign, GPT-4o reaches wF1 20.76 on emotion from video, AffectGPT collapses to “neutral” (11.03), and a hearing annotator scores wAF 21.39 on 7-class sentiment [6].]

[TODO: P4 – our approach in one paragraph.]

Contributions.

- 1) [TODO: C1: factorized non-manual encoder separating grammatical from affective function, with an explicit disentanglement metric rather than an assertion.]
- 2) [TODO: C2: a manual prosody channel for affect, quantified by ablation.]

- 3) [TODO: C3: state of the art on EmoSign under signer-independent LOSO, at orders of magnitude fewer parameters and less memory than MLLM baselines.]
- 4) [TODO: C4: end-to-end emotion-aware pipeline deployable on a 4GB consumer GPU, with a full accuracy/latency/VRAM Pareto analysis.]

II. RELATED WORK

A. Isolated Sign Language Recognition

[TODO: ST-GCN lineage [7]; SAM-SLR/SL-GCN [8]; SPOTER [9]; sub-1M-parameter skeleton models [10]; training-strategy gains [11]; RGB upper bounds [12]. End with our position: we adopt compact recipes and claim no ISLR novelty.]

B. Sign Language Translation

[TODO: Neural SLT and PHOENIX14T [13]; gloss-free line [14]–[17]; compact SLT [18]; YouTube-ASL T5-base scaling [19]; the caution that reported gains often stem from data rather than architecture [20].]

C. Emotion in Sign Language

[TODO: EmoSign [6]; eJSL [21]; DGS with facial action units [22]; emotion in sign language conversation [23]. Note the field is 2025–2026 and nascent; nobody treats the grammatical/affective confound as an explicit training objective.]

D. Sign Language Production and Avatars

[TODO: SignLLM [24]; SignAvatar [25]; MediaPipe’s 52 ARKit-compatible blendshapes [26]. Our position: deterministic retargeting with emotion-parametric modulation; generative motion is future work.]

TABLE I

DATASETS USED. SPLITS ARE SIGNER-INDEPENDENT THROUGHOUT;
EMOSIGN USES 4-FOLD LEAVE-ONE-SIGNER-OUT CROSS-VALIDATION.

Dataset	Lang.	Clips	Signers	Role
ASL Citizen [27]	ASL	83,399	52	ISLR training
WLASL-100/300 [28]	ASL	[TODO: n]	—	Comparability
ASLLRP [29]	ASL	2,651	19	Video + L labels
EmoSign [6]	ASL	200	4	Affect benchmark
How2Sign [30]	ASL	~35k	11	Continuous SLT
DFEW [31]	—	16,000	—	NM pretraining
MAFW [32]	—	10,045	—	NM pretraining
INCLUDE [33]	ISL	4,287	7	Cross-lingual baseline

[TODO: Fig. 3 – qualitative: wh-question / negation clips with neutral affect. Entangled baseline predicts anger/frustration; factorized model does not. This is the most persuasive figure.]

Fig. 2. Grammatical markers misread as affect by the entangled baseline.
[TODO: One sentence per term explaining why it exists. λ_4 use gradient reversal [35] so neither factor can predict the other's label; λ_6 penalises feature overlap; λ_7 minimises a contrastive log-ratio upper bound on mutual information [36].]

[TODO: Fig. 1 – pipeline architecture]

Fig. 1. SEAM pipeline. Keypoints and 52 blendshapes feed three channels (manual, non-manual, prosody); the non-manual stream is factorized into a linguistic factor z_L and an affective factor z_A , which condition text generation and avatar animation.

III. DATASETS

[TODO: Prose: why ASL Citizen over WLASL (first-party hosting, no link rot, IRB consent); EmoSign's inter-annotator ceiling (mean Krippendorff $\alpha = 0.593$; sentiment 0.738, joy 0.699, worry 0.555, surprise– 0.119, disgust 0.166) and what it bounds; the key methodological point that EmoSign is built on ASLLRP so the same utterances carry both grammatical-marker and affect labels; How2Sign keypoints only; rejected datasets and why (PHOENIX14T is affectively flat; YouTube-ASL quality unverified).]

IV. METHOD

A. Perception and Preprocessing

[TODO: MediaPipe Tasks [26]: 543 landmarks, 52 ARKit-compatible blendshapes, head pose, gaze. One Euro Filter smoothing [34]. 12 fps sampling, $T = 64$ windows; justify 12 fps by the $O(n^2)$ attention argument and the published 0.5 BLEU-4 cost [18].]

B. Three Channels

[TODO: Manual M (85 keypoints, 255-dim/frame); non-manual NM (52 blendshapes, AU proxies, head dynamics, gaze); prosody P (speed, amplitude, repetition, pause, jerk). Justify P from the literature: EmoSign annotators named sign size, speed and repetition as cues [6], and hand motion improves signer-emotion recognition [21].]

C. Factorized Non-Manual Encoder

Let $z_L = \text{Enc}_L(NM)$ and $z_A = \text{Enc}_A(NM, P)$. We train

$$\begin{aligned} \mathcal{L} = & \lambda_1 \text{CE}(h_L(z_L), y_{\text{mark}}) + \lambda_2 \text{CE}(h_A(z_A), y_{\text{emo}}) \\ & + \lambda_3 \text{MSE}(h_{VA}(z_A), y_{VA}) \\ & + \lambda_4 \text{CE}(h'_A(\text{GRL}(z_L)), y_{\text{emo}}) + \lambda_5 \text{CE}(h'_L(\text{GRL}(z_A)), y_{\text{mark}}) \\ & + \lambda_6 \|z_L^\top z_A\|_F^2 + \lambda_7 \text{vCLUB}(z_L; z_A). \end{aligned} \quad (1)$$

D. Measuring Disentanglement

[TODO: Frozen cross-probes: predict emotion from z_L and markers from z_A . Report cross-prediction AUC, where 0.5 denotes perfect separation. Define this before showing results.]

E. Emotion-Conditioned Generation

[TODO: Control tokens derived from z_A injected into T5-small [37]. Disclose here that the emotion-styled paraphrase corpus is LLM-generated and semantically filtered.]

F. Avatar Control

[TODO: Bone retargeting with joint clamping and quaternion continuity; 52 blendshapes mapped to VRM/ARKit expression targets; z_A modulates blendshape gain, movement amplitude and timing, motivated by ASL affective prosody [4].]

G. Efficiency by Design

[TODO: State that the system is lightweight by design, not by post-hoc compression: keypoints rather than pixels, 12 fps rather than 24, $\leq 2\text{M}$ -parameter recognition, T5-small, INT8 export.]

V. EXPERIMENTAL SETUP

[TODO: Training on RTX 4060 16 GB and free cloud; inference measured on RTX 3050 4 GB, contrasted with the 80 GB A100 used for the published MLLM baselines [6]. AdaFactor, constant lr 1×10^{-3} , 10-epoch warmup, grad-clip 1.0, label smoothing 0.1, effective batch 128, beam 5, 256-frame input / 128-token output caps. Seeds and repetitions. SacreBLEU [38] default tokenization; BERTScore [39]; weighted accuracy and weighted F1. Restate the LOSO protocol. State that every reported number comes from a logged run.]

TABLE II
ISOLATED RECOGNITION. SIGNER-INDEPENDENT SPLITS. LATENCY
MEASURED ON AN RTX 3050.

Dataset	Model	Params	Top-1	Top-5	msConfiguration	Cross-AUC ↓	Emo wF1 ↑	Marker F1 ↑	
WLASL-100	GRU	XX.X	XX.X	XX.X	XX.X	Single branch (entangled)	XX.X	XX.X	XX.X
	ST-GCN	XX.X	XX.X	XX.X	XX.X	Two branch, no separation loss	XX.X	XX.X	XX.X
	Pose Transformer	XX.X	XX.X	XX.X	XX.X	+ GRL	XX.X	XX.X	XX.X
ASL Citizen	GRU	XX.X	XX.X	XX.X	XX.X	+ orthogonality	XX.X	XX.X	XX.X
	ST-GCN	XX.X	XX.X	XX.X	XX.X	+ vCLUB (full)	XX.X	XX.X	XX.X
	Pose Transformer	XX.X	XX.X	XX.X	XX.X				

TABLE III
EMOSIGN AFFECT RECOGNITION. BASELINE ROWS ARE QUOTED FROM
[6] (EVALUATED ON AN 80 GB A100); OUR ROWS ARE MEAN ± STD OVER
4-FOLD LEAVE-ONE-SIGNER-OUT CROSS-VALIDATION. **[TODO:
RE-CHECK EVERY QUOTED BASELINE CELL AGAINST THE PUBLISHED
TABLES 3 AND 4 OF [6] BEFORE SUBMISSION.]**

Modality	Model	Sent-7 wF1	Emo wF1	Params
video	MiniGPT4-video	13.03	22.02	7B+
video	Qwen2.5-VL-7B	[TODO: verify]	18.53	7B
video	AffectGPT	—	11.03	7B+
video	GPT-4o	26.35	20.76	—
video	Hearing person	21.39	—	—
video	SEAM (ours)	XX.X	XX.X	XX.X
video+cap	GPT-4o	—	55.09	—
video+cap	AffectGPT	—	47.77	7B+
video+cap	SEAM (ours)	XX.X	XX.X	XX.X

VI. RESULTS

- A. Isolated Recognition
- B. Affect Recognition versus Frontier MLLMs
- C. Disentanglement
- D. Prosody and Pretraining Ablations
- E. Emotion-Conditioned Generation
- F. Continuous Translation
- G. Efficiency

VII. SYSTEM AND DEMO

[TODO: FastAPI + WebSocket runtime, React Three Fiber + three-vm viewer, measured latency HUD, avatar screenshots under four emotions, demo video link. Note the deliberate localhost-only, unauthenticated-by-default posture for webcam-derived biometric data.]

VIII. LIMITATIONS AND ETHICS

[TODO: Do not soften this section. Cover: 200 clips / 4 signers and high variance even under LOSO; the annotation ceiling ($\bar{\alpha} = 0.593$, surprise— 0.119, disgust 0.166) bounding per-class reliability; whether L supervision used ASLLRP annotations or the pseudo-label fallback; disclosure of the LLM-generated paraphrase corpus; domain shift from lab-recorded scripted utterances; the level of Deaf-community involvement in our work, stated honestly; no clinical, legal or emergency-setting readiness claim;

TABLE IV
DISENTANGLEMENT ABLATION. CROSS-PREDICTION AUC: 0.5 DENOTES
PERFECT SEPARATION. 4-FOLD LOSO, MEAN ± STD.

TABLE V
CHANNEL AND PRETRAINING ABLATIONS. 4-FOLD LOSO ON EMOSIGN.

Configuration	Sent-7 wAF	Emo wF1
Non-manual only, from scratch	XX.X	XX.X
+ prosody channel	XX.X	XX.X
+ DFEW/MAFW pretraining	XX.X	XX.X
+ both (full)	XX.X	XX.X

data stewardship (no video redistribution — we release code, weights and labels only); and which INCLUDE categories were available if the cross-lingual ablation ran on a partial download.]

IX. CONCLUSION

[TODO: Restate the factorization insight and the two headline numbers. Name future work concretely: continuous emotion-aware translation, generative expressive motion, ISL/NSL extension, and larger Deaf-annotated affect corpora.]

ACKNOWLEDGMENTS

[TODO: Deaf annotators and consultants, dataset providers, compute providers.]

REFERENCES

- [1] C. Valli and C. Lucas, *Linguistics of American Sign Language: An Introduction*. Gallaudet University Press, 2000.
- [2] R. Pfau, M. Steinbach, and B. Woll, *Sign Language*. De Gruyter Mouton, 2012.
- [3] E. A. Elliott and A. M. Jacobs, “Facial expressions, emotions, and sign languages,” *Frontiers in Psychology*, vol. 4, p. 115, 2013.
- [4] J. S. Reilly, M. L. McIntire, and H. Seago, “Affective prosody in american sign language,” *Sign Language Studies*, vol. 75, no. 1, pp. 113–128, 1992.
- [5] J. K. Hietanen, J. M. Leppänen, and U. Lehtonen, “Perception of emotions in the hand movement quality of finnish sign language,” *Journal of Nonverbal Behavior*, vol. 28, pp. 53–64, 2004.
- [6] P. Chua, C. M. Fang, T. Ohkawa, R. Kushalnagar, S. Nanayakkara, and P. Maes, “Emosign: A multimodal dataset for understanding emotions in american sign language,” *arXiv preprint arXiv:2505.17090*, 2025, 200 ASL utterances from ASLLRP; 4 signers; 3 Deaf annotators; 7-point sentiment + 10 emotion intensities + free-text cues; mean Krippendorff alpha 0.593.
- [7] S. Yan, Y. Xiong, and D. Lin, “Spatial temporal graph convolutional networks for skeleton-based action recognition,” in *AAAI Conference on Artificial Intelligence*, 2018.
- [8] S. Jiang, B. Sun, L. Wang, Y. Bai, K. Li, and Y. Fu, “Skeleton aware multi-modal sign language recognition,” *arXiv preprint arXiv:2103.08833*, 2021, 98.42% RGB / 98.53% RGB-D on AUTSL, 2021 ChaLearn signer-independent challenge.

TABLE VI

EMOTION-CONDITIONED GENERATION. STYLE ACCURACY FROM A HELD-OUT CLASSIFIER; SEMANTIC PRESERVATION AS BERTSCORE-F1 AGAINST THE NEUTRAL REFERENCE; HUMAN PREFERENCE IS BLINDED PAIRWISE.

Condition	Style acc.	BERTScore-F1	Human pref.
Neutral (reference)	—	1.00	—
Happy	XX.X	XX.X	XX.X
Sad	XX.X	XX.X	XX.X
Angry	XX.X	XX.X	XX.X

TABLE VII

GLOSS-FREE CONTINUOUS TRANSLATION ON HOW2SIGN. BASELINES QUOTED FROM [18], [19].

System	Params	fps	BLEU-4
T5-base [19]	248M	12–30	11.89
Compact SLT [18]	77M	24	10.06
Compact SLT [18]	77M	12	9.53
SEAM (ours)	XX.X	24	XX.X
SEAM (ours)	XX.X	12	XX.X

- [9] M. Boháček and M. Hruší, “Sign pose-based transformer for word-level sign language recognition,” in *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2022, pp. 182–191.
- [10] T. verify author list, “Signbart: New approach with the skeleton sequence for isolated sign language recognition,” *arXiv preprint arXiv:2506.21592*, 2025, 749,888 parameters; 96.04% on LSA-64.
- [11] —, “Training strategies for isolated sign language recognition,” *arXiv preprint arXiv:2412.11553*, 2024, +1.63% WLASL, +14.12% Slovo over prior best.
- [12] —, “Self-supervised video transformers for isolated sign language recognition,” *arXiv preprint arXiv:2309.02450*, 2023, maskFeat, 79.02% top-1 on gloss-based WLASL2000.
- [13] N. C. Camgöz, S. Hadfield, O. Koller, H. Ney, and R. Bowden, “Neural sign language translation,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018.
- [14] B. Zhou, Z. Chen, A. Clapés, J. Wan, Y. Liang, S. Escalera, Z. Lei, and D. Zhang, “Gloss-free sign language translation: Improving from visual-language pretraining,” in *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023, arXiv:2307.14768.
- [15] K. Lin, X. Wang, L. Zhu, K. Sun, B. Zhang, and Y. Yang, “Gloss-free end-to-end sign language translation,” in *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL)*, 2023, arXiv:2305.12876.
- [16] R. Wong, N. C. Camgöz, and R. Bowden, “Sign2gpt: Leveraging large language models for gloss-free sign language translation,” in *International Conference on Learning Representations (ICLR)*, 2024, arXiv:2405.04164.
- [17] H. Liang, C. Huang, Y. Xu, C. Tang, W. Ye, J. Zhang, X. Chen, J. Yu, and L. Xu, “Llava-slt: Visual language tuning for sign language translation,” *arXiv preprint arXiv:2412.16524*, 2024.
- [18] K. Chen and M. Tsai, “Towards compact sign language translation: Frame rate and model size trade-offs,” *arXiv preprint arXiv:2605.09554*, 2026, 77M params: MMPose 85 keypoints (255-dim) - ζ linear - ζ T5-small; BLEU-4 10.06 @24fps vs 9.53 @12fps on How2Sign; T5-base 248M reaches 11.89.
- [19] D. Uthus, G. Tanzer, and M. Georg, “Youtube-asl: A large-scale, open-domain american sign language-english parallel corpus,” in *Advances in Neural Information Processing Systems*, vol. 36, 2023, pp. 29029–29047.
- [20] T. verify author list, “Gloss-free sign language translation: An unbiased evaluation of progress in the field,” *arXiv preprint arXiv:2603.13240*, 2026.
- [21] —, “Emotion recognition in signers,” *arXiv preprint arXiv:2512.15376*, 2026, eJSL: Japanese Sign Language; 2 signers x

TABLE VIII

END-TO-END EFFICIENCY MEASURED ON AN RTX 3050 (4 GB). PEAK VRAM IS THE MAXIMUM OBSERVED ACROSS THE FULL PIPELINE.

Config	Precision	fps	p50 (ms)	p95 (ms)	Peak VRAM (MB)
Full	FP32	XX.X	XX.X	XX.X	XX.X
Full	INT8	XX.X	XX.X	XX.X	XX.X
12 fps	INT8	XX.X	XX.X	XX.X	XX.X

78 utterances x 7 emotions = 1092 clips; finds hand motion improves signer-emotion recognition.

- [22] —, “Emotion recognition in german sign language with facial action units,” in *Workshop on the Representation and Processing of Sign Languages*, 2026.
- [23] —, “Emotion recognition in sign language conversation,” *arXiv preprint arXiv:2605.23328*, 2026.
- [24] S. Fang, L. Wang, C. Zheng, Y. Tian, and C. Chen, “Signllm: Sign language production large language models,” *arXiv preprint arXiv:2405.10718*, 2024.
- [25] T. verify author list, “Signavatar: Sign language 3d motion reconstruction and generation,” *arXiv preprint arXiv:2405.07974*, 2024.
- [26] C. Lugaresi, J. Tang, H. Nash, C. McClanahan, E. Ubaweja, M. Hays, F. Zhang, C.-L. Chang, M. G. Yong, J. Lee *et al.*, “Mediapipe: A framework for building perception pipelines,” *arXiv preprint arXiv:1906.08172*, 2019, tasks API FaceLandmarker emits 52 ARKit-compatible blendshape coefficients.
- [27] A. Desai, L. Berger, F. Minakov *et al.*, “Asl citizen: A community-sourced dataset for advancing isolated sign language recognition,” in *Advances in Neural Information Processing Systems (Datasets and Benchmarks)*, 2023, 83,399 videos; 2,731 signs; 52 signers; arXiv:2304.05934.
- [28] D. Li, C. Rodriguez, X. Yu, and H. Li, “Word-level deep sign language recognition from video: A new large-scale dataset and methods comparison,” in *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2020.
- [29] C. Neidle, A. Opoku, and D. Metaxas, “Asl video corpora and sign bank: Resources available through the american sign language linguistic research project (asllrp),” *arXiv preprint arXiv:2201.07899*, 2022, source corpus for EmoSign; includes gloss, English text and linguistic non-manual annotations.
- [30] A. Duarte, S. Palaskar, L. Ventura, D. Ghadiyaram, K. DeHaan, F. Metze, J. Torres, and X. Giro-i Nieto, “How2sign: A large-scale multimodal dataset for continuous american sign language,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021.
- [31] X. Jiang, Y. Zong, W. Zheng, C. Tang, W. Xia, C. Lu, and J. Liu, “Dfaw: A large-scale database for recognizing dynamic facial expressions in the wild,” in *Proceedings of the 28th ACM International Conference on Multimedia*, 2020, pp. 2881–2889.
- [32] Y. Liu, W. Dai, C. Feng, W. Wang, G. Yin, J. Zeng, and S. Shan, “Mafw: A large-scale, multi-modal, compound affective database for dynamic facial expression recognition in the wild,” in *Proceedings of the 30th ACM International Conference on Multimedia*, 2022, pp. 24–32.
- [33] A. Sridhar, R. G. Ganesan, P. Kumar, and M. Khapra, “Include: A large scale dataset for indian sign language recognition,” in *Proceedings of the 28th ACM International Conference on Multimedia*, 2020, 4,287 videos; 263 word signs; 15 word categories; 0.27M frames.
- [34] G. Casiez, N. Roussel, and D. Vogel, “1 euro filter: A simple speed-based low-pass filter for noisy input in interactive systems,” in *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI)*, 2012, pp. 2527–2530.
- [35] Y. Ganin and V. Lempitsky, “Unsupervised domain adaptation by backpropagation,” in *International Conference on Machine Learning (ICML)*, 2015, gradient reversal layer.
- [36] P. Cheng, W. Hao, S. Dai, J. Liu, Z. Gan, and L. Carin, “Club: A contrastive log-ratio upper bound of mutual information,” in *International Conference on Machine Learning (ICML)*, 2020.
- [37] C. Raffel, N. Shazeer, A. Roberts, K. Lee, S. Narang, M. Matena, Y. Zhou, W. Li, and P. J. Liu, “Exploring the limits of transfer learning with a unified text-to-text transformer,” *Journal of Machine Learning Research*, vol. 21, no. 140, pp. 1–67, 2020.

- [38] M. Post, “A call for clarity in reporting bleu scores,” in *Proceedings of the Third Conference on Machine Translation: Research Papers*, 2018, pp. 186–191.
- [39] T. Zhang, V. Kishore, F. Wu, K. Q. Weinberger, and Y. Artzi, “Bertscore: Evaluating text generation with bert,” in *International Conference on Learning Representations (ICLR)*, 2020.